CLINICAL · LARYNX & AIRWAY

The hoarse child and the child who cannot swallow

Children's voices and swallows fail for different reasons from adults', and they cannot always tell you what is wrong. A hoarse child is usually a shouting child with nodules, but the second commonest cause is recurrent respiratory papillomatosis, which can obstruct the airway. A child who coughs with feeds may have a neurological swallow, a laryngeal cleft, a tracheo-oesophageal fistula or a nose that is too narrow to breathe through while sucking. Both problems are assessed with the same tools, flexible endoscopy and instrumental swallow studies, and managed by the same partnership of ENT surgeon and speech and language therapist.

AT A GLANCE

- **Nodules** are the commonest cause of a hoarse child and respond to voice therapy; **RRP** (HPV 6 and 11) is second.
- **RRP:** debulking; cidofovir or bevacizumab for frequent recurrence; HPV vaccination.
- **Vocal fold paralysis** follows about 5% of aortic arch operations; injection or ansa–RLN reinnervation if persistent.
- **FEES and videofluoroscopy** show penetration (to the folds) and aspiration (below them), often silent in children.
- **Laryngeal clefts** (Benjamin–Inglis I–IV): palpate the interarytenoid area at every microlaryngoscopy.

1 CHAPTER ONE The developing larynx

Why children differ.

1.1 Anatomy with age

The infant larynx sits high, at about the level of C2 to C3, so the epiglottis can meet the soft palate and the infant can breathe while sucking. It descends through childhood to about C6 to C7 by fifteen. The infant larynx is about a third of adult size, the subglottis (bounded by the complete cricoid ring) is the narrowest point, and the vocal folds are short with an immature layered structure that matures through childhood and puberty. Babies breathe preferentially through the nose for

about the first six months, so nasal obstruction disrupts the coordinated suck, swallow and breathe cycle. Mature chewing and the ability to manage most textures develop by about two years, following motor development more than age.



FIGURE 1 Laryngomalacia: an omega-shaped epiglottis, a reminder of how the infant larynx differs.

Dhingra, fig. 59.1

2 CHAPTER TWO The hoarse child

Voice disorders.

2.1 Causes

Dysphonia affects between 4 and 23% of children at some stage, and most causes are benign and self-limiting.

CAUSES OF DYSPHONIA IN CHILDREN

GROUP	CAUSES	POINTS
Behavioural	**Vocal fold nodules**	Commonest cause; boys, loud voice use; voice therapy works well

GROUP	CAUSES	POINTS
Infective	**Recurrent respiratory papillomatosis**	Second commonest; HPV 6 and 11; may threaten the airway
Structural	Cysts, sulcus, polyps	Stroboscopy shows a disturbed mucosal wave
Congenital	Webs, laryngeal cleft, subglottic stenosis, cysts	From birth; often with stridor or feeding problems
Neurological	Vocal fold paralysis	Often after cardiac surgery; Arnold–Chiari malformation if bilateral
Inflammatory	Reflux, eosinophilic oesophagitis	Arytenoid oedema and erythema
Iatrogenic	Intubation injury, webs after repeated surgery	Granuloma, scar, posterior glottic stenosis
Functional	Puberphonia, functional dysphonia	Normal larynx, abnormal use

2.2 Assessment

Listen to the voice for pitch, range, projection, breaks and resonance, and ask about voice use, noisy breathing, feeding, cardiac surgery and intubation. Look for syndromic features. A paediatric flexible nasendoscope (as small as 2.1 mm) shows mobility and gross lesions in most children; rigid stroboscopy is tolerated by mature children from about three or four years and shows the mucosal wave. **Laryngeal ultrasound** is a useful, non-invasive way to assess vocal fold movement in uncooperative children. A paralysed fold needs imaging of the nerve's course, and a contrast swallow or oesophagoscopy is considered when reflux or eosinophilic oesophagitis is suspected.

2.3 Vocal fold nodules

Nodules are bilateral swellings at the junction of the anterior and middle thirds, more common in boys, and caused by voice overuse. They respond well to **voice therapy**, often regress at puberty in boys, and are almost never operated on in children.

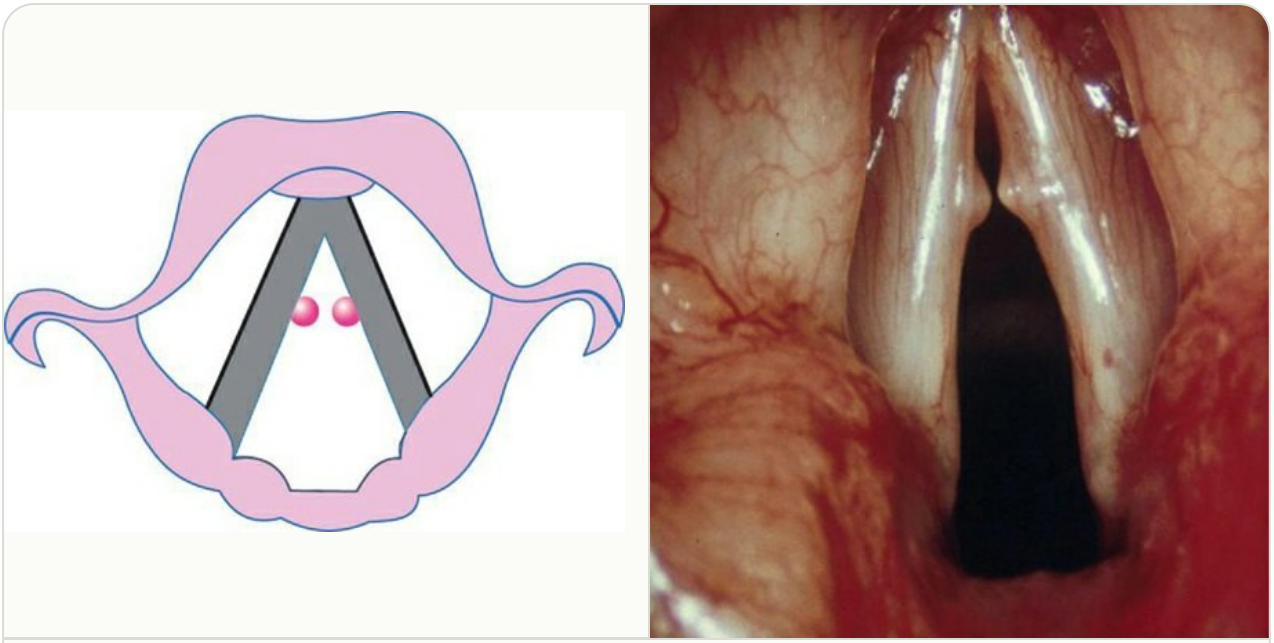


FIGURE 2 The site of vocal nodules, and nodules on both folds.
Dhingra, figs 61.1 and 61.2

2.4 Recurrent respiratory papillomatosis

RRP is caused by HPV types 6 and 11, usually acquired at birth from the mother's genital tract; juvenile-onset disease typically appears before five years with progressive hoarseness and later stridor, and can be mistaken for asthma or croup. Treatment is repeated **microdebrider or cold-steel debulking**, preserving normal tissue and avoiding the anterior commissure to prevent webs; KTP laser is an alternative. Tracheostomy is avoided where possible because it may spread disease distally. Adjuvant treatments for aggressive disease (more than about four procedures a year) include intralesional **cidofovir**, whose small cancer risk must be discussed with parents, and systemic **bevacizumab**, which has given striking responses. **HPV vaccination** prevents the disease in the next generation and may reduce recurrences.

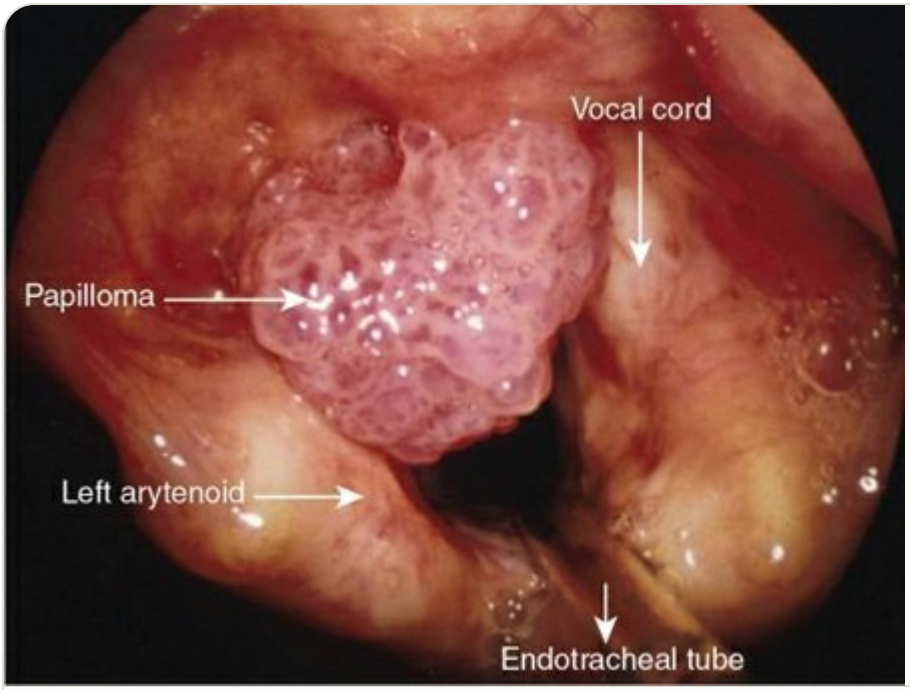


FIGURE 3 Laryngeal papillomatosis.
Dhingra, fig. 61.11

2.5 Vocal fold paralysis

Paralysis is the second commonest congenital laryngeal anomaly. Unilateral paralysis, often left-sided after cardiac or aortic arch surgery (about 5% of such operations), causes a weak cry, feeding problems and aspiration; children having arch surgery should be screened before discharge. Bilateral paralysis causes stridor with a good cry and is associated with Arnold–Chiari malformation and hydrocephalus. Many idiopathic palsies recover, so observation is appropriate when breathing and feeding are safe. Persistent unilateral palsy is treated with **injection medialisation** (temporary materials such as Gelfoam for about six weeks or hyaluronic acid for months, and longer-lasting calcium hydroxylapatite or fat), which in children needs a general anaesthetic and overnight oximetry, or later with thyroplasty or **ansa cervicalis to recurrent laryngeal nerve reinnervation**, which gives good results in children.

2.6 Webs and other lesions

Anterior glottic webs may be congenital, often with a subglottic extension and associated with 22q11.2 deletion, or acquired after repeated microlaryngoscopy. Thin webs are divided endoscopically with a keel or mitomycin; thick congenital

webs may need an open repair. Phonosurgery in children follows adult principles but is more conservative: cold instruments, no laser on the true folds, and subepithelial infusion to protect the cover.

3 CHAPTER THREE The child who cannot swallow safely

Dysphagia.

3.1 Assessment

Swallowing problems affect about 1% of children and many more in clinical groups such as prematurity, neurodisability and cardiac disease. A feeding history (breast or bottle, textures, duration of meals, coughing, colour change, vomiting, chest infections, weight gain) and observation of a meal by a speech and language therapist come first. Instrumental tests are **videofluoroscopy**, which shows every phase including oesophageal transit, and **FEES**, which can be done at the bedside or during breastfeeding, shows secretions and laryngeal sensation, and avoids radiation; the two are comparable for the pharyngeal phase. They show *penetration* (material in the laryngeal vestibule down to the vocal folds) and *aspiration* (material below the folds), which in children is often silent.

3.2 Causes

CAUSES OF PAEDIATRIC DYSPHAGIA

LEVEL	CAUSES
Nose and nasopharynx	Choanal atresia, piriform aperture stenosis, mid-nasal stenosis, craniosynostosis, adenoids
Mouth and oropharynx	Cleft palate, Pierre Robin sequence, macroglossia, tongue-tie (rarely), tonsillar hypertrophy
Larynx and pharynx	Laryngomalacia, vocal fold paralysis, laryngeal cleft, cricopharyngeal dysfunction
Oesophagus and trachea	Oesophageal atresia and tracheo-oesophageal fistula, vascular rings, strictures, eosinophilic oesophagitis

LEVEL	CAUSES
Neurological	Cerebral palsy, hypoxic brain injury, Chiari malformation, neuromuscular disease
Respiratory and cardiac	Chronic lung disease of prematurity, congenital heart disease (breathlessness, fatigue)

3.3 Laryngeal cleft

A posterior laryngeal cleft is a failure of fusion between the larynx and oesophagus. In the **Benjamin–Inglis** classification, *type I* is confined to the interarytenoid area above the vocal folds; *type II* extends into the cricoid lamina; *type III* extends through the cricoid into the cervical trachea; and *type IV* extends into the thoracic trachea. Type I clefts present with coughing on feeds, chest infections and a weak or hoarse cry, and are easily missed unless the interarytenoid area is **palpated** at microlaryngoscopy. Thickened feeds help many; persistent aspiration is treated by **injection augmentation** of the interarytenoid area or endoscopic repair. Deeper clefts present earlier and more severely and need endoscopic or open repair, with tracheostomy and gastrostomy for type III and IV.

3.4 Other important causes

Oesophageal atresia with tracheo-oesophageal fistula presents at birth with drooling, choking and inability to pass a nasogastric tube, or later with an isolated H-type fistula and coughing on feeds. **Vascular rings** compress the trachea and oesophagus, causing stridor and dysphagia for solids. **Eosinophilic oesophagitis** causes food refusal, vomiting and food impaction in older children. **Tongue-tie** can interfere with breastfeeding and is divided by simple frenotomy when it clearly does, but it is not a common cause of feeding failure.

3.5 Management

Management is multidisciplinary: positioning, pacing, bottle teats and texture modification by the speech and language therapist, dietetic support, treatment of reflux, and non-oral feeding by nasogastric tube or **gastrostomy** when the swallow

is unsafe or intake inadequate, sometimes with fundoplication. The structural cause is corrected where possible.

THE CHILD WITH A VOICE OR SWALLOWING PROBLEM

1. **Airway first**
   - → **stridor or distress** airway assessment and microlaryngoscopy and bronchoscopy
   - → **safe** step 2
2. **Voice**
   - → flexible endoscopy or ultrasound
   - → **nodules** voice therapy
   - → **papillomas** debulking, adjuvant therapy if frequent
   - → **paralysis** image the nerve course, observe, inject or reinnervate
3. **Swallow**
   - → feeding history and observed meal
   - → videofluoroscopy or FEES
   - → **aspiration** thickened feeds, non-oral feeding if unsafe

4 **Look for structure**

- → interarytenoid palpation for cleft
- → contrast study for fistula, ring or stricture
- → endoscopy and biopsy for eosinophilic oesophagitis
- → nasal endoscopy for nasal causes

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Why the child larynx differs:** position, size, subglottis.
- **Nodules and papillomatosis:** the two commonest causes of a hoarse child.
- **RRP adjuvants:** cidofovir and bevacizumab; HPV vaccination.
- **Vocal fold paralysis after cardiac surgery;** injection medialisation and reinnervation in children.
- **Penetration versus aspiration;** FEES versus videofluoroscopy.
- **Benjamin–Inglis classification** of laryngeal clefts; palpate the interarytenoid area.

4.2 Common pitfalls

COMMON PITFALLS

- Treating papillomatosis as asthma or croup.
- Missing a type I laryngeal cleft by not palpating the interarytenoid area.
- Operating on vocal fold nodules in a child.

- Lasering the anterior commissure and creating a web.

Written for this app from Oxford Textbook of Otolaryngology ch. 79, 80, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

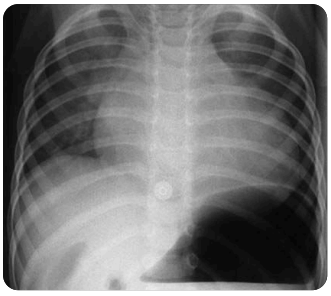
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



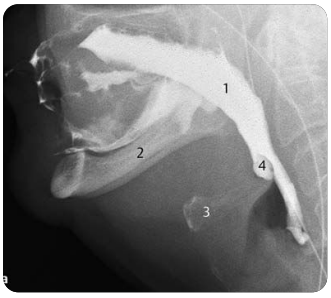
CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Aspiration

- Occasionally aspiration is caused by mechanical failure of the epiglottis to close because of the presence of tumor, but most cases are due to neurologic (sensory) disorders without any visible pathology apart from the aspiration. Beware of aspiration pneumonia.
- The above-mentioned pathologies are partly demonstrated in other sections.



First stage: the barium contrast (1) just reaches the esophagus. For orientation note the mandible (2), hyoid (3), and vallecula (4)….

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

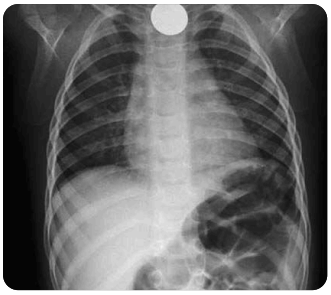
- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…